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Clinical Utility of LC-MS/MS for Blood Myo-Inositol in Patients with Acute Kidney Injury and Chronic Kidney Disease

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BACKGROUND: Diagnosing acute kidney injury (AKI) and chronic kidney disease (CKD) relies on creatinine, which lacks optimal diagnostic sensitivity. The kidney-specific proximal tubular enzyme myo-inositol oxygenase (MIOX) catalyzes the conversion of myo-inositol (MI) to D-glucuronic acid. We hypothesized that proximal tubular damage, which occurs in AKI and CKD, will decrease MIOX activity, causing MI accumulation. To explore this, we developed an LC-MS/MS assay to quantify plasma MI and assessed its potential in identifying AKI and CKD patients.

METHODS: MI was quantified in plasma from 3 patient cohorts [normal kidney function (n = 105), CKD (n = 94), and AKI (n = 54)]. The correlations between MI and creatinine were determined using Deming regression and Pearson correlation and the impact of age, sex, and ethnicity on MI concentrations was assessed. Receiver operating characteristic curve analysis was employed to evaluate MI diagnostic performance.

RESULTS: In volunteers with normal kidney function, the central 95th percentile range of plasma MI concentrations was 16.6 to 44.2 µM. Age, ethnicity, and sex showed minimal influence on MI. Patients with AKI and CKD exhibited higher median MI concentrations [71.1 (25th percentile: 38.2, 75th percentile: 115.4) and 102.4 (77, 139.5) µM], respectively. MI exhibited excellent sensitivity (98.9%) and specificity (100%) for diagnosing CKD. In patients with AKI, MI increased 32.9 (SD 16.8) h before creatinine.

CONCLUSIONS: This study unveils MI as a potential renal biomarker, notably elevated in plasma during AKI and CKD. Plasma MI rises 33 h prior to serum creatinine, enabling early AKI detection. Further validation and exploration of MI quantitation in kidney disease diagnosis is warranted.

Introduction

Kidney disease is a significant cause of death and disability globally (1, 2). In acute kidney injury (AKI), abnormalities in the structure or function of the kidney result in profound pathophysiological changes (3). Chronic kidney disease (CKD) is characterized by progressive kidney function decline, highlighted by persistently low estimated glomerular filtration (eGFR) and albuminuria (4). Creatinine is widely used to assess kidney function in patients with known or suspected kidney disease but has significant limitations (3, 5).

Creatinine is a byproduct of creatine metabolism, The circulatory concentration is affected by muscle mass and pathologies that impact tubular secretion, reabsorption, and by extension, creatinine clearance (6, 7). Because creatinine is produced at a constant rate and there is significant kidney functional reserve, its elevation post-AKI is gradual, resulting in a prolonged time to diagnosis (6). In critical care settings, 18% to 50% of intensive care unit patients develop AKI, contributing to high mortality and morbidity (8–10). Creatinine may fail to rapidly identify patients with AKI and does not discriminate between intrinsic structural damage and functional loss (11–13). Lack of early and specific AKI diagnosis has been cited as a major reason why new therapeutics have failed in clinical trials (14, 15). There is a pressing need for biomarkers that offer greater sensitivity and responsiveness to real-time shifts in kidney structural damage and function.

Myo-inositol oxygenase (MIOX), a 32-kDa cytosolic enzyme, is expressed specifically in renal proximal tubules (5, 16). MIOX catalyzes the conversion of myo-inositol (MI) into D-glucuronic acid and plays an important role in maintaining kidney redox balance.

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Overexpression of MIOX in mice is associated with exacerbation of cisplatin-induced AKI (7). We and others have previously demonstrated that MIOX concentrations change following kidney injury, preceding creatinine in a murine AKI model and in critically ill AKI patients (5, 7, 17).

There is limited literature addressing the clinical utility of MI, the substrate of MIOX, for evaluating the presence or absence of renal injury. MI, a cyclohexane containing 6 hydroxyl groups, constitutes 1 of 9 stereoisomers within the inositol family (18). MI is physiologically relevant, with important roles in cellular metabolism and human reproduction (18). Proximal tubular damage releases MIOX into plasma, potentially limiting the ability to metabolize MI (5). Similarly, CKD results in proximal tubular atrophy, potentially limiting MIOX function. With acute and chronic proximal tubular damage and diminished MIOX activity, MI may increase in plasma and serve as a diagnostic indicator of kidney injury.

This study reports the development of a LC-MS/MS assay for quantifying MI in plasma and investigates the utility of plasma MI to discriminate patients with normal kidney function from those with AKI and CKD.

Materials and Methods

STANDARDS AND CHEMICALS

The materials utilized for the preparation of calibration standards, internal standards, and quality control (QC) consisted of the following: MI (Sigma-Aldrich), D6-myo-inositol (Sigma-Aldrich), acetonitrile (Sigma-Aldrich), bovine serum albumin (30%, Sigma-Aldrich), and formic acid (Sigma-Aldrich). Purified water was obtained from the Milli-Q IQ 15 Ultrapure & Pure Lab Water Purification System with a resistivity of 18 MΩ (Millipore Sigma).

CALIBRATORS, INTERNAL STANDARDS, AND QUALITY CONTROL SAMPLES

Five calibration solutions spanning 0.1 to 1000 µM were prepared in a 5% albumin/0.9% saline matrix. D6-MI served as an internal standard. QC samples were composed of pooled human plasma at endogenous concentrations (low QC) and pooled human plasma at endogenous concentrations supplemented with 100 µM MI (high QC).

LC-MS/MS CONDITIONS

The LC-MS/MS system utilized components from Waters, including an ACQUITY I-class UPLC interfaced to a XEVO TQ-S mass spectrometer. Chromatography employed a BEH Z-HILIC column (2.1 × 100 mm, 1.7 µm) maintained at 40°C. The mobile phases consisted of 0.1% formic acid in water (mobile phase A) and 0.1% formic acid in acetonitrile (mobile phase B). Injection volume was 10 µL. A discontinuous elution gradient was employed starting at 5% mobile phase and a flow rate of 0.6 mL/min to 0.8 min, ramping to 50% mobile phase A at 5 min. Mobile phase A was then increased to 80% at a flow rate of 0.4 mL/min to 9.5 min followed by a return to 5% A at 0.6 mL/min and re-equilibration from 10 to 13 min. MI exhibited a retention time of 3.92 min (Table 1).

The XEVO TQS mass spectrometer was operated in multiple reaction monitoring mode using electrospray ionization with negative ion mode. Mass transitions were 86 and 117 for MI and 89 and 102 for D6-MI (Table 1).

Operating conditions included a desolvation temperature of 600°C, desolvation gas flow of 1000 L/h, cone gas flow of 150 L/h, and nebulizer gas flow of 7.0 bar. Dwell time was 25 ms. Mass transitions were monitored for 25 ms dwell time.

To assess ion suppression, alternating injections were made using 10 µM MI in water and fetal bovine serum, alongside 100 µM MI in water and fetal bovine serum.

SAMPLE PREPARATION

Samples for analysis were prepared by mixing 100 µL of plasma, 50 µL of internal standard, and 600 µL of acetonitrile in microcentrifuge tubes, which were then vortexed for 5 s and centrifuged for 7 min at 3000 *g*. Supernatants were transferred into LC-MS/MS autosampler vials for testing.

METHOD VALIDATION

The limit of quantitation was determined using serial 10-fold dilutions of 10 mM MI in a 5% albumin in saline, which was assayed in duplicate. Intra-assay and interassay variability was assessed using 2 samples in 2 replicates on 14 different days. The stability of MI in heparin and EDTA plasma is well established (19, 20). Future studies investigating matrix effects will be important to fully validate the method.

SPECIMENS

The institutional review board at the Washington University School of Medicine approved this study (IRB# 201305087). Heparin plasma samples were obtained from the Barnes Jewish Hospital Core Laboratory. The cohort of volunteers with normal kidney function (healthy cohort) were seen as outpatients with normal serum creatinine and eGFR (>60 mL/min/1.73 m^2). Samples were included that also had normal urinalysis, urine albumin-to-creatinine

Table 1. LC-MS/MS validation outcomes.						
Compound name	Retention time (min)	Precursor ion (m/z)	Product ion	ESI mode	Cone voltage (V)	Collision energy (V)
MI	3.92	179	86	Negative	54	16
MI	3.92	179	177	Negative	36	12
D6-MI	3.92	185	89	Negative	54	16
D6-MI	3.92	185	102	Negative	54	20

Abbreviation: ESI, electrospray ionization.

ratio, or protein-to-creatinine ratio when available. CKD cohort samples were sourced from patients clinically diagnosed with CKD but who had not started dialysis. Specimens were selected based solely on creatinine concentrations ≥2.5 mg/dL (221 umol/L), following which CKD diagnosis and clinical stage were confirmed from the electronic medical record.

Specimens for the AKI cohort were identified using custom software implemented in the Python3 programming language. Briefly, a daily automated query retrieved all new creatinine results from the Laboratory Information System (Cerner Millenium, Oracle) and all previous creatinine results from the same patients. Patients were automatically classified as AKI based solely on changes in serum creatinine in accordance with the Kidney Disease Improving Global Outcomes (KDIGO) guidelines (3). AKI stage 1 is defined as having either an absolute serum creatinine increase of >0.3 mg/dL from baseline or a 1.5- to 1.9-fold increase from baseline. Urine output was not used to determine AKI status. Finally, a web application built using Dash presented creatinine trends for each patient meeting KDIGO criteria and facilitated manual review and selection of samples. The source code for this software is available at https://doi.org/10.6084/m9.figshare.25943929.v1. Samples preceding specimens that displayed notable increases in creatinine concentrations were designated as baseline samples, i.e., sample 0 h (n = 54 AKI patients). When available, between 1 and 4 remnant samples collected prior to sample 0 h were obtained. Two to 4 post-sample 0 h specimens were also obtained. Samples were initially stored at 4°C in the clinical laboratory for 4 to 8 days to allow for add-on testing, following which they were sent to the research lab where they were aliquoted and frozen at −80°C. All samples were analyzed within 5 weeks from the date of collection and were stored for <4 weeks at −80°C. MI is stable in plasma for up to 14 days at 4°C (21). Patient demographics for all cohorts are listed in Supplemental Table 1.

DATA ANALYSIS

LC-MS/MS data were acquired on Masslynx v4.2 and processed with Targetlynx XS v4.2. Patient data were analyzed and graphed using the GraphPad Prism Software, v9.1.9. Outliers were identified using the robust regression and outlier removal method on GraphPad Prism (22). Normality was assessed on GraphPad using the Kolmogorov–Smirnov and Shapiro–Wilk normality tests (23). Skewness and kurtosis were also assessed, and data was deemed to be normally distributed if outputs were between −1 and +1. Group comparisons were performed using the Mann–Whitney test unless specified. P values <0.05 were considered statistically significant. Deming regression, Pearson correlation, and receiver operating characteristic curve analyses were also utilized. Optimal thresholds for distinguishing MI concentrations in CKD patients from healthy controls were identified using the Youden Index. The average time to MI increase in the AKI cohort was calculated by averaging the time it took for MI to increase relative to baseline based on the difference in blood draw times between the baseline and subsequent samples. Similarly, the time to reach AKI stage 1 by serum creatine (sCr) criteria was determined using the difference between baseline (i.e., pre-AKI, time 0 samples) and samples with sCr rising from baseline (i.e., Pre-AKI, Time 0 h samples) to AKI stage 1.

Results

ASSAY CHARACTERISTICS

Total imprecision (CV) was 3.90% at 34.3 µM and 5.62% at 128 µM. The limit of blank and limit of detection were calculated to be 0.5 µM and 1 µM, respectively. The assay was linear from 1 to 1000 µM.

MI CONCENTRATION IN HEALTHY PATIENTS

MI concentrations were normally distributed in our healthy cohort. We found little to no significant difference in MI concentration with age (Fig. 1A), sex (Fig. 1B), or ethnicity (Fig. 1C). Although weak, the correlation between MI concentration and age is statistically significant ($P = 0.025$), suggesting a minimal impact of age on MI concentrations. The reference range, representing the 2.5% to 97.5% percentile of the cohort, was 16.4 to 44.3 µM (Fig. 1D). Notably, in this cohort,

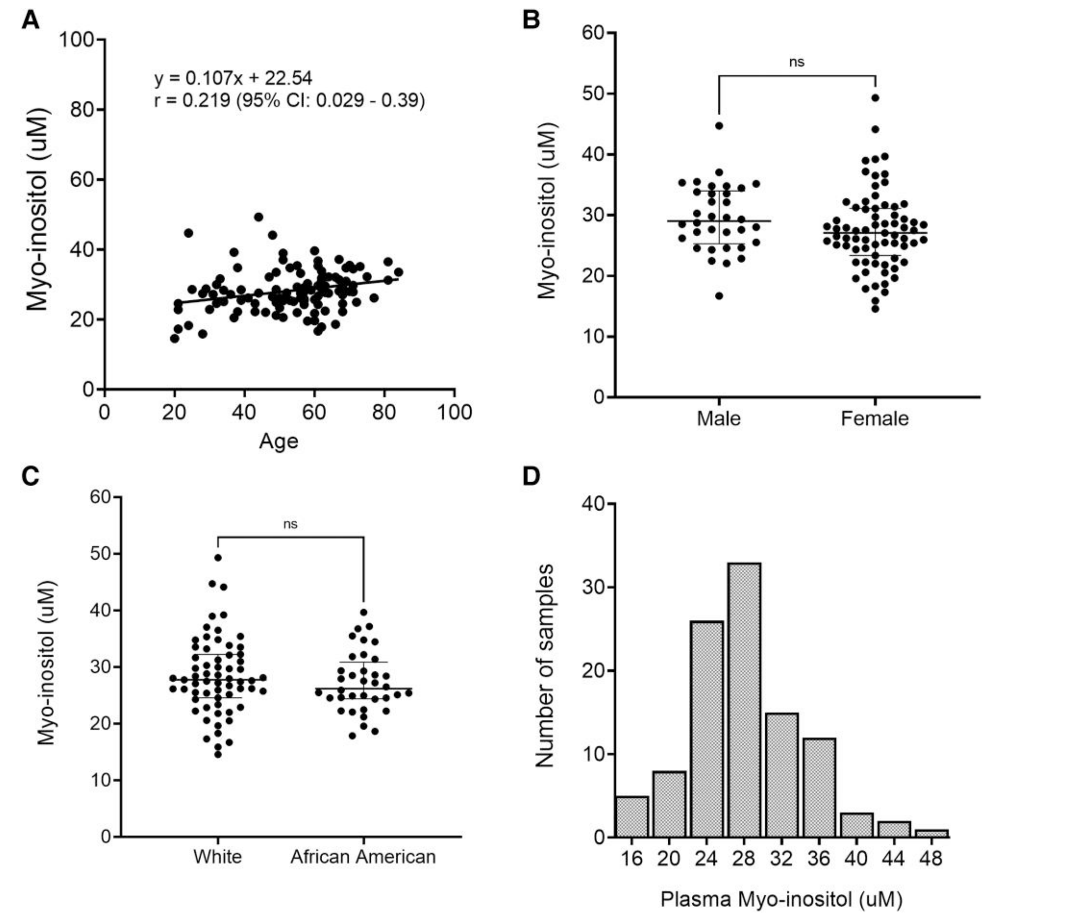

Fig. 1. MI concentrations across sex, age, and ethnicity in a healthy patient cohort. (A), Deming regression analyses comparing MI concentrations with age; (B), MI concentration between sexes (female = 71; male = 34); (C), MI concentrations between ethnicities (Black = 36; non-Hispanic White = 63); (D), 95th percentile distribution of MI in healthy cohort. N = 105.

creatinine concentrations were independent of age and ethnicity but differed with sex ((Fig. 2A–C).

PLASMA MI IS ELEVATED IN PATIENTS WITH CKD

In 94 patients with clinically diagnosed CKD who were yet to initiate dialysis, MI concentrations were higher relative to the healthy cohort [Fig. 3A, median (MI) = 102.4 µM (95% CI: 89.91–113.1)], although MI concentrations were not normally distributed in this cohort. Receiver operating characteristic curve analyses revealed an area under the receiver operating characteristic (AUROC) curve of 0.9997 (95% CI: 0.9989–1.00), and an MI cutoff of 50.22 µM had a diagnostic sensitivity of 98.94% (95% CI: 94.22–99.95%) and specificity of 100% (95% CI: 96.47%–100%) for distinguishing between healthy individuals and CKD patients. AUROC for sCr was 1.0 (95% CI: 1.0–1.0) for corresponding sCr values.

PLASMA MI INCREASES BEFORE CREATININE IN AKI

In 54 patients with creatinine-defining AKI per KDIGO criteria, MI increased on average 32.88 (95% CI: 28.3–37.5) h before creatinine (Fig. 4A, $P < 0.0001$). At this pre-AKI stage, median MI concentrations were significantly higher at 40.02 µM [25th percentile (Q1): 27.85, 75th percentile (Q3): 68.7] compared to the median MI concentration in the healthy cohort [27.64 µM (Q1: 24.57, Q3: 31.98), $P < 0.0001$]. At AKI stage 1 (time 32.88 h), the median concentration of MI in patients was 71.10 µM (Q1: 38.22, Q3: 115.4) after an outlier was removed. Of the 54 patients identified with baseline sCr with AKI (stage 1), 26 developed stage 2, and 10 developed stage 3 AKI. The median MI concentration for

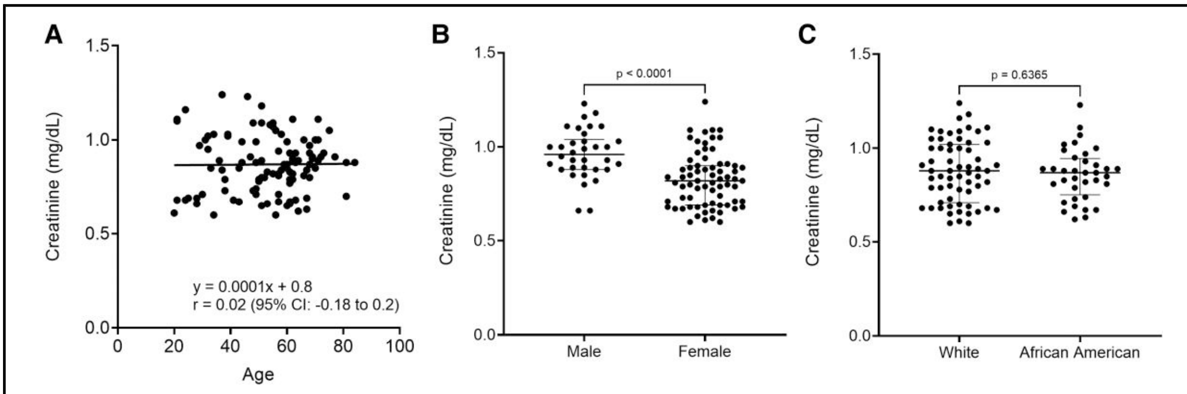

Fig. 2. Creatinine concentrations across sex, age, and ethnicity in the healthy patient cohort. (A), Deming regression analyses comparing creatinine with age; (B), creatinine concentrations between sex (female = 71; male = 34); (C), creatinine concentrations between racial groups (Black = 36; non-Hispanic White = 63). 1 mg/dL sCr = 0.0884 mmol/L.

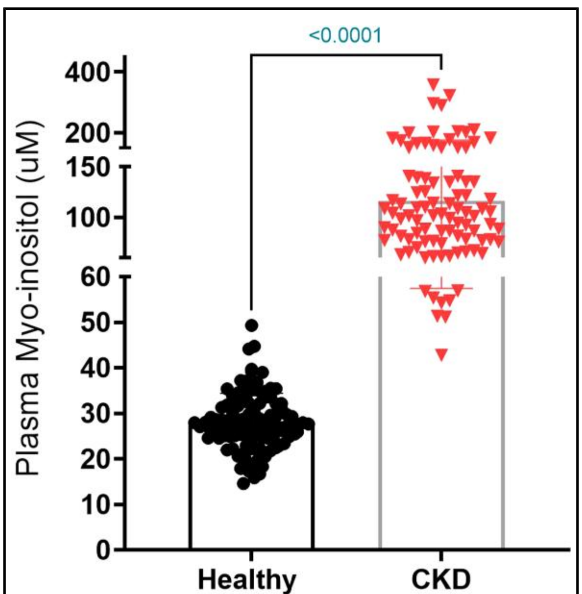

Fig. 3. Comparing MI in the healthy cohort relative to the CKD cohort. Concentrations of MI in the healthy cohort vs the CKD cohort. n = 105 healthy patients, n = 94 CKD patients. Color figure available at https://academic.oup.com/clinchem.

stage 2 was 86.20 (Q1: 39.80, Q3: 196.5) µM, and stage 3 was 173.2 (Q1: 70.13, Q3: 260) µM.

In contrast, no statistically significant difference in creatinine was observed between the healthy cohort and the pre-AKI cohort, $P = 0.117$, although at the AKI stage creatinine concentrations were significantly higher than the pre-AKI stage as expected, $P < 0.0001$ (Fig. 4B).

At the pre-AKI stage, a cutoff MI concentration of 26.5 µM exhibited an AUROC of 0.76 (0.67–0.85), sensitivity of 81.5% (69.2–89.6%), and specificity of 41.9% (32.9–51.5%) (Fig. 4C). Also, 25 out of the 54 patients in this cohort exhibited MI concentrations greater than the upper reference limit (i.e., >44.3 µM) at the pre-AKI stage. An AUROC of 0.58 (0.47–0.69) was observed for corresponding sCr values.

Concentrations of MI at the AKI stage displayed an AUROC of 0.90 (0.83–0.96) and a sensitivity of 92.6% (82.5–97.1%) and a specificity of 73.3% (64.2–80.9%) at a cutoff of 31.41 µM (Fig. 4D). An AUROC of 0.97 (0.94–0.997) was observed for corresponding sCr values.

MI RELATIONSHIP WITH CREATININE

Relative to creatinine, MI concentrations demonstrated an overall slope of 0.01 (95% CI: 0.006–0.015) and intercept of 0.57 (95% CI: 0.45–0.70) with a Pearson r of 0.43 (95% CI: 0.25–0.57) in the healthy cohort (Fig. 5A). In the AKI cohort, Deming regression analyses of MI relative to creatinine revealed a slope of 0.002 (95% CI: 0.001–0.004) and a y-intercept of −1.34 (95% CI: 1.16–1.52) (Fig. 5B). MI was poorly correlated with creatinine (Pearson $r = 0.35$, 95% CI: 0.086–0.56). Whereas in the CKD cohort, a slope of 0.01 (95% CI: 0.006–0.014), y-intercept of 2.86 (95% CI: 2.41–3.32), and r of 0.52 (95% CI: 0.35–0.65) (Fig. 5C) was observed.

Discussion

There is a pressing need for sensitive and precise biomarkers to assess renal integrity and function, as current

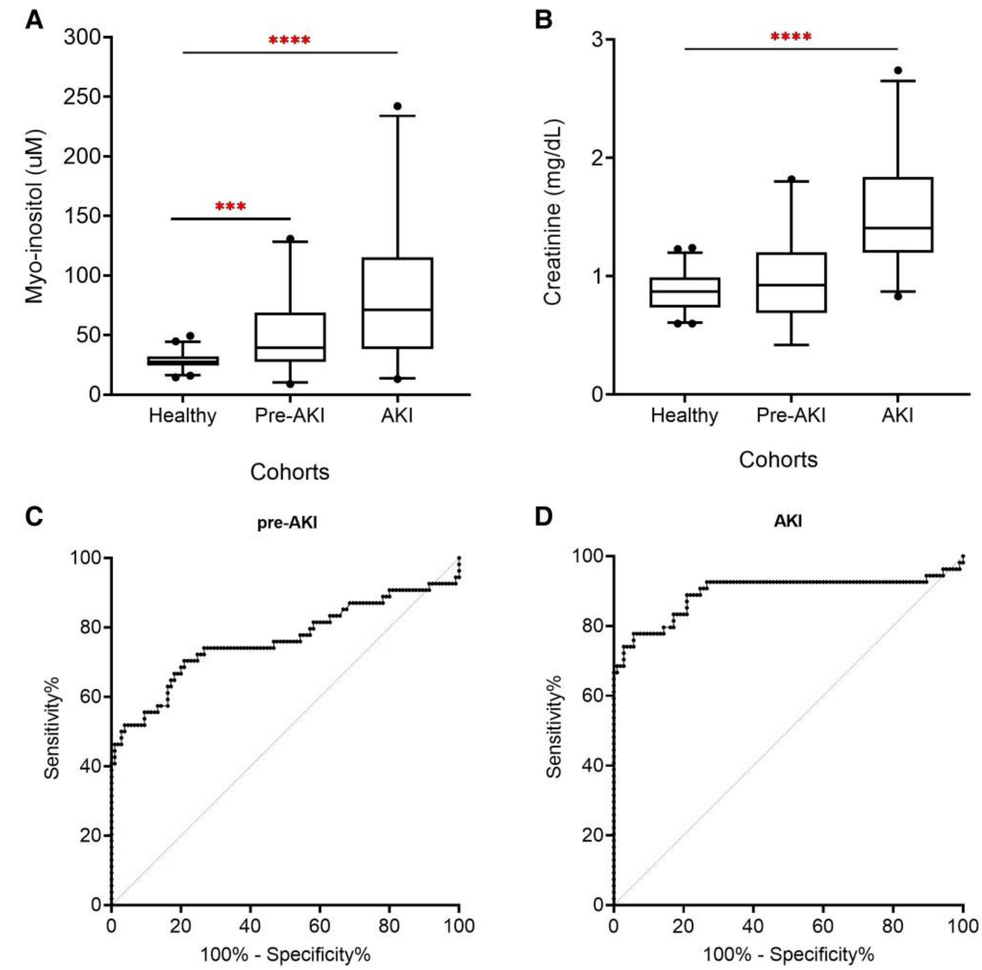

Fig. 4. MI concentrations are increased 33 h before creatinine concentrations. (A), MI concentrations in patient cohorts with normal kidney function, pre-AKI, and AKI; (B), creatinine concentrations in patient cohorts with normal kidney function, pre-AKI, and AKI; (C), receiver operating characteristic curve analyses of MI concentrations at the pre-AKI stage; (D), receiver operating characteristic curve analyses of MI concentrations at the AKI stage. Whiskers in A and B represent 2.5th to 97.5th percentiles. One outlier was identified and removed from the pre-AKI cohorts in panel A. 1 mg/dL sCr = 0.0884 mmol/L. Color figure available at https://academic.oup.com/clinchem.

clinical guidelines primarily rely on changes in serum creatinine concentrations and urine output, which have limitations (3). Creatinine concentration, commonly used in these assessments, is influenced by external nonrenal factors and may not distinguish between functional loss and structural damage (6, 12, 13, 24). In 2015, we introduced MIOX as a potential plasma biomarker for AKI diagnosis, and subsequent evidence suggests its prognostic value in the assessment of both AKI and CKD (2, 5, 25). However, further investigations into MIOX's role in kidney disease management have been hindered by the scarcity of commercial assays for quantifying MIOX.

Our study focused on developing an LC-MS/MS-based assay to quantify MI, the substrate for MIOX. Evidence indicates that MIOX may be released from the proximal tubules into plasma and urine in the presence of kidney injury or atrophy (5). With less MIOX available for metabolism of MI, we hypothesized that MI would correspondingly accumulate in plasma. This investigation aimed to explore MI's potential as a diagnostic tool for AKI and CKD. Herein, we demonstrate a robust assay for MI that is able to accurately differentiate between healthy patients and those with CKD and AKI. Furthermore, MI increases before creatinine in patients developing AKI, predicting AKI with high

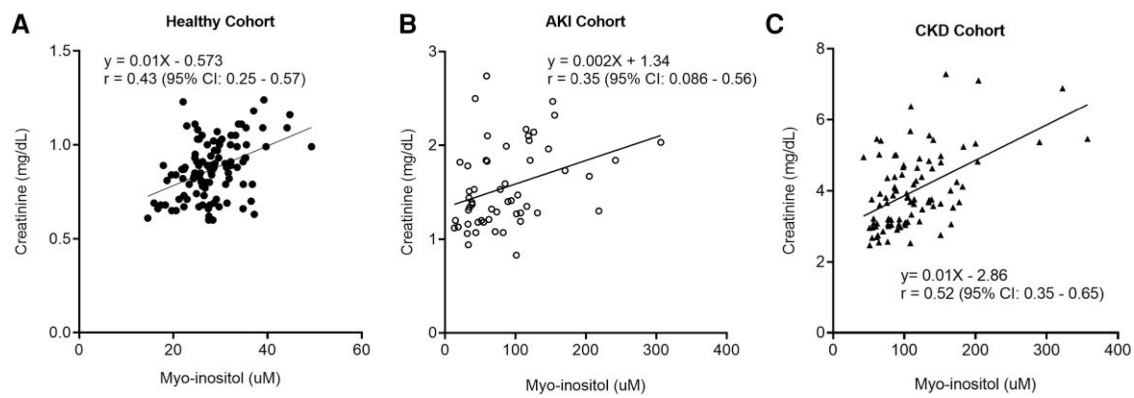

Fig. 5. Deming regression and Pearson correlation analyses comparing MI to creatinine concentrations in the (A) healthy cohort, (B) AKI cohort, and (C) CKD cohort. Three outliers for creatinine were identified (22) and removed before Deming regression and Pearson correlation analyses in the CKD cohort in panel C. 1 mg/dL sCr = 0.0884 mmol/L.

sensitivity and specificity. The kinetics of MI differ from creatinine, suggesting a potential nonredundant role in kidney disease testing. These findings underscore the promise of MI as a valuable biomarker for assessing renal function and integrity.

The limitations of creatinine in promptly and accurately identifying early kidney damage pose challenges for timely AKI detection. This delayed diagnosis can increase the risk of progression into CKD and lead to higher mortality rates (8, 13). Given that a significant proportion of hospitalized patients are vulnerable to AKI, there is an urgent need for timely diagnoses and interventions to halt AKI progression and prevent the onset of end-stage kidney disease (8, 26). Therefore, identifying a biomarker whose concentrations are altered before creatinine concentrations is crucial.

In our AKI cohort, MI concentrations exhibited a significant surge, with an average lead time of 32.9 h (SD: 16.8 h) before creatinine-dependent AKI became evident. Notably, MI showed substantial elevation in pre-AKI samples, even in the presence of normal creatinine levels and inadequate creatinine deltas for AKI diagnoses according to KDIGO guidelines (3). Moreover, MI levels remained elevated during the AKI stage. These findings suggest the potential superiority of MI in terms of sensitivity and specificity for early AKI diagnosis compared to creatinine, although further studies are necessary to validate this observation. The observed performance characteristics align with those of other novel and promising kidney biomarkers, including neutrophil gelatinase-associated lipocalin, kidney injury molecule-1, as well as tissue inhibitor metalloproteinase-2 and insulin-like growth factor-binding protein 7 (26–29). Interestingly, in 5 patients at the AKI stage, MI was reduced relative to the pre-AKI stage, suggesting different kinetics in response to injury and possible recovery compared to creatinine. This observation adds a nuanced understanding to MI's dynamics in the context of kidney injury and recovery.

Several extrarenal factors impact creatinine concentrations. The influence of race on creatinine levels is a subject of ongoing debate. Studies conducted in the general US population suggest that Black individuals tend to display higher average serum creatinine concentrations, likely linked to a higher prevalence of kidney diseases within this demographic (30). Consequently, historical practices have involved using race-modified creatinine-based equations for estimating kidney function (31). However, it has been demonstrated that these equations tend to overestimate eGFR in patients of African ancestry with compromised kidney function (32, 33). Recognizing the detrimental effects on patient care, there has been a growing movement to eliminate the use of race as a factor in creatinine-based eGFR calculations (34, 35).

It is worth noting that MI concentrations are disrupted in various conditions such as cognitive diseases, diabetes, and polycystic ovarian syndrome (36–39). Numerous studies have underscored the influence of MI supplementation on these conditions (36, 38, 40). Future investigations into the clinical effectiveness of MI as a marker of kidney injury must therefore consider confounding by other coexisting conditions and potential interference from exogenous MI intake.

In our study, we examined MI in healthy cohorts of Black and non-Hispanic White patients, finding equivalent concentrations between the 2 populations. This suggests that MI may serve as a valuable secondary measure of kidney function unaffected by racial differences. If confirmed, this could contribute to promoting more equitable healthcare practices across diverse populations. Furthermore, MI concentrations were only modestly influenced by sex or age, resembling the performance of cystatin C, another renal biomarker less affected by such variables (41). Notably, in our healthy patient cohorts, creatinine was not influenced by race. We also did not observe a strong correlation between creatinine and age in contrast to other published studies (42, 43). Our cohort comprised participants aged 20 years or older, with almost 70% aged 50 years and older. This observation may be due to the mixed-sex correlation analysis, as males are known to have more significant decline in sCr with age relative to females (44). Moreover, after 70 years of age, correlations between age and sCr are absent (45).

In this study, we also observed that MI had excellent sensitivity (98%) and specificity (100%) for identifying CKD at a cutoff of 50 µM, well beyond the upper reference limit in our healthy cohort. Additionally, the correlation between creatinine and MI was modest among our patient cohorts (0.35, 0.43, and 0.52 for the AKI, healthy, and CKD cohorts, respectively). These findings suggest that MI serves as a nonredundant biomarker relative to creatinine, offering potential for additional clinical and biological insights that may be specific to etiology. Supporting this notion, a previous study demonstrated that elevated levels of MIOX, the parent enzyme of MI, at hospital discharge following recovery from community-acquired AKI are associated with higher risks of developing CKD (2). Further investigations are required to fully elucidate the relationship between creatinine and MI in kidney disease.

The study has several limitations. First, AKI cohorts were identified using pre-established creatinine criteria, and CKD cohorts were initially identified based on creatinine, with clinical stage confirmation through electronic medical record reviews. Thus, further investigations incorporating urine output criteria for AKI diagnoses according to KDIGO guidelines are essential to validate our findings. Second, the reliance on creatinine in clinical AKI and CKD diagnoses introduces it as a confounding variable in our study, which seeks to challenge its suitability as the ideal primary endpoint for kidney disease diagnoses (26). Future studies incorporating measured GFR may add clarity to the utility of MI in CKD evaluation. Third, the causes of injury in our CKD and AKI cohort were not specified, highlighting the need for a more in-depth exploration of MI concentrations in relation to different AKI etiologies. Fourth, the AKI cohort was composed of remnant clinical samples identified retrospectively based on stringent inclusion criteria. A prospective study that identifies patients at risk of developing AKI and monitors them with serial measurements throughout the development of AKI should be undertaken, which may influence the sensitivity and specificity of MI for AKI. Fifth, this study was conducted at a single center and lacked representation of many major racial/ethnic groups within the United States. This limitation reflects the patient demographics served by our institution. To address this, a larger-scale study with a more diverse patient population is warranted.

Despite the limitations of our study, there are several notable strengths. Importantly, our research stands as the first to assess plasma MI concentrations as a renal biomarker, employing an LC-MS/MS-based assay with high analytic reproducibility and variability. Additionally, the use of the substrate of a kidney-localized enzyme to indicate kidney injury adds a unique dimension compared to traditional creatinine-based assessment (26). Furthermore, our study encompassed a substantial number of patient specimens. Nonetheless, for comprehensive validation, it is imperative to conduct further studies with diverse cohorts and emerging kidney biomarkers.

In summary, our findings highlight that MI exhibits distinct performance characteristics and kinetics when compared to creatinine, suggesting its potential as an independent marker for assessing kidney function and integrity. Notably, in patients with AKI, MI concentrations show elevation more than a day before observable increases in creatinine, indicating potential higher sensitivity in detecting early changes in kidney function. However, it is essential to conduct additional studies that explore the kinetics of MI in patients with comorbid conditions such as diabetes and cardiovascular disease. Moreover, assessing the impact of disease etiology on MI concentrations will provide a more comprehensive understanding of its utility as a renal biomarker.

Supplemental Material

Supplemental material is available at *Clinical Chemistry* online.

Nonstandard Abbreviations: AKI, acute kidney injury; CKD, chronic kidney disease; MIOX, myo-inositol oxygenase; MI, myo-inositol; Q1, 25th percentile; Q3, 75th percentile; eGFR, estimated glomerular filtration rate; QC, quality control; KDIGO, Kidney Disease Improving Global Outcomes; sCr, serum creatine; AUROC, area under the receiver operating characteristic.

Author Contributions: *The corresponding author takes full responsibility that all authors on this publication have met the following required criteria of eligibility for authorship: (a) significant contributions to the conception and design, acquisition of data, or analysis and interpretation of data; (b) drafting or revising the article for intellectual content; (c) final approval of the published article; and (d) agreement to be accountable for all aspects of the article thus ensuring that questions related to the accuracy* *or integrity of any part of the article are appropriately investigated and resolved. Nobody who qualifies for authorship has been omitted from the list.*

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References

1. Chertow GM, Burdick E, Honour M, Bonventre JV, Bates DW. Acute kidney injury, mortality, length of stay, and costs in hospitalized patients. J Am Soc Nephrol 2005;16:3365–70.
2. Kakkanattu TJ, Kaur J, Nagesh V, Kundu M, Kamboj K, Kaur P, et al. Serum myo-inositol oxygenase levels at hospital discharge predict progression to chronic kidney disease in community-acquired acute kidney injury. Sci Rep 2022;12:13225.
3. Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. Kidney Inter Suppl 2012;2:1.
4. KDIGO. CKD evaluation and management. https://kdigo.org/guidelines/ckd-evaluation-and-management/ (Accessed April 2023).
5. Gaut JP, Crimmins DL, Ohlendorf MF, Lockwood CM, Griest TA, Brada NA, et al. Development of an immunoassay for the kidney specific protein myo-inositol oxygenase, a potential biomarker of acute kidney injury. Clin Chem 2014;60:747–57.
6. McMahon GM, Waikar SS. Biomarkers in nephrology. Am J Kidney Dis 2013;62: 165–78.
7. Dutta RK, Kondeti VK, Sharma I, Chandel NS, Quaggin SE, Kanwar YS. Beneficial effects of myo-inositol oxygenase deficiency in cisplatin-induced AKI. J Am Soc Nephrol 2017;28:1421–36.
8. Mo S, Bjelland TW, Nilsen TIL, Klepstad P. Acute kidney injury in intensive care patients: incidence, time course, and risk factors. Acta Anaesthesiol Scand 2022;66:961–8.
9. Schanz M, Schöffski O, Kimmel M, Oberacker T, Göbel N, Franke UFW, et al. Under-recognition of acute kidney injury after cardiac surgery in the ICU impedes early detection and prevention. Kidney Blood Press Res 2021;47:50–60.
10. White KC, Serpa-Neto A, Hurford R, Clement P, Laupland KB, See E, et al. Sepsis-associated acute kidney injury in the intensive care unit: incidence, patient characteristics, timing, trajectory, treatment, and associated outcomes. A multicenter, observational study. Intensive Care Med 2023;49:1079–89.
11. Himmelfarb J, Joannidis M, Molitoris B, Schietz M, Okusa MD, Warnock D, et al. Evaluation and initial management of acute kidney injury. Clin J Am Soc Nephrol 2008;3:962–7.
12. El-Khoury JM, Hoenig MP, Jones GRD, Lamb EJ, Parikh CR, Tolan NV, et al. AACC guidance document on laboratory investigation of acute kidney injury. J Appl Lab Med 2021;6:1316–37.
13. Molitoris BA. Low-flow acute kidney injury: the pathophysiology of prerenal azotemia, abdominal compartment syndrome, and obstructive uropathy. Clin J Am Soc Nephrol 2022;17:1039–49.
14. Vaara ST, Bhatraju PK, Stanski NL, McMahon BA, Liu K, Joannidis M, et al. Subphenotypes in acute kidney injury: a narrative review. Crit Care 2022;26:251.
15. Birkelo BC, Pannu N, Siew ED. Overview of diagnostic criteria and epidemiology of acute kidney injury and acute kidney disease in the critically ill patient. Clin J Am Soc Nephrol 2022;17:717–35.
16. Charalampous FC, Lyras C. Biochemical studies on inositol. IV. Conversion of inositol to glucuronic acid by rat kidney extracts. J Biol Chem 1957;228:1–13.
17. Sharma I, Deng F, Liao Y, Kanwar YS. Myo-inositol oxygenase (MIOX) overexpression drives the progression of renal tubulointerstitial injury in diabetes. Diabetes 2020;69:1248–63.
18. Russo M, Forte G, Montanino Oliva M, Laganà AS, Unfer V. Melatonin and myo-inositol: supporting reproduction from the oocyte to birth. Int J Mol Sci 2021;22:8433.
19. Schimpf KJ, Meek CC, Leff RD, Phelps DL, Schmitz DJ, Cordle CT. Quantification of myo-inositol, 1,5-anhydro-D-sorbitol, and D-chiro-inositol using high-performance liquid chromatography with electrochemical detection in very small volume clinical samples. Biomed Chromatogr 2015;29: 1629–36.
20. Leung K-Y, Mills K, Burren KA, Copp AJ, Greene NDE. Quantitative analysis of myo-inositol in urine, blood and nutritional supplements by high-performance liquid chromatography tandem mass spectrometry. J Chromatogr B Analyt Technol Biomed Life Sci 2011;879:2759–63.
21. Ward RM, Sweeley J, Lugo RA. Inositol analysis by HPLC and its stability in scavenged sample conditions. Med Chem (Los Angeles) 2015;5:077–80.
22. Motulsky HJ, Brown RE. Detecting outliers when fitting data with nonlinear regression—a new method based on robust nonlinear regression and the false discovery rate. BMC Bioinformatics 2006;7:123.
23. Mishra P, Pandey CM, Singh U, Gupta A, Sahu C, Keshri A. Descriptive statistics

and normality tests for statistical data. Ann Card Anaesth 2019;22:67–72.
24. Barai S, Gambhir S, Prasad N, Sharma RK, Ora M. Functional renal reserve capacity in different stages of chronic kidney disease. Nephrology 2010;15:350–3.
25. Mertoglu C, Gunay M, Gurel A, Gungor M. Myo-inositol oxygenase as a novel marker in the diagnosis of acute kidney injury. J Med Biochem 2018;37:1–6.
26. van Duijl TT, Ruhaak LR, de Fijter JW, Cobbaert CM. Kidney injury biomarkers in an academic hospital setting: where are we now? Clin Biochem Rev 2019;40:79–97.
27. Mishra J, Dent C, Tarabishi R, Mitsnefes MM, Ma Q, Kelly C, et al. Neutrophil gelatinase-associated lipocalin (NGAL) as a biomarker for acute renal injury after cardiac surgery. Lancet 2005;365:1231–8.
28. Dent CL, Ma Q, Dastrala S, Bennett M, Mitsnefes MM, Barasch J, et al. Plasma neutrophil gelatinase-associated lipocalin predicts acute kidney injury, morbidity and mortality after pediatric cardiac surgery: a prospective uncontrolled cohort study. Crit Care Lond Engl 2007;11:R127.
29. Edelstein CL. Biomarkers of acute kidney injury. Adv Chronic Kidney Dis 2008;15:222–34.
30. Peralta CA, Risch N, Lin F, Shlipak MG, Reiner A, Ziv E, et al. The association of African ancestry and elevated creatinine in the coronary artery risk development in young adults (CARDIA) study. Am J Nephrol 2010;31:202–8.
31. Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. Kidney Int Suppl 2013;3:1–150.
32. Fabian J, Kalyesubula R, Mkandawire J, Hansen CH, Nitsch D, Musenge E, et al. Measurement of kidney function in Malawi, South Africa, and Uganda: a multicentre cohort study. Lancet Glob Health 2022;10:e1159–69.
33. Levey AS, Titan SM, Powe NR, Coresh J, Inker LA. Kidney disease, race, and GFR estimation. Clin J Am Soc Nephrol 2020;15:1203–12.
34. Delgado C, Baweja M, Crews DC, Eneanya ND, Gadegbeku CA, Inker LA, et al. A unifying approach for GFR estimation: recommendations of the NKF-ASN task force on reassessing the inclusion of race in diagnosing kidney disease. Am J Kidney Dis 2022;79:268–288.e1.
35. Pierre CC, Marzinke MA, Ahmed SB, Collister D, Colón-Franco JM, Hoenig MP, et al. AACC/NKF guidance document on improving equity in chronic kidney disease care. J Appl Lab Med 2023;8:789–816.
36. Greene DA, De Jesus PV, Winegrad AI. Effects of insulin and dietary myoinositol on impaired peripheral motor nerve conduction velocity in acute streptozotocin diabetes. J Clin Invest 1975;55:1326–36.
37. Shimon H, Agam G, Belmaker RH, Hyde TM, Kleinman JE. Reduced frontal cortex inositol levels in postmortem brain of suicide victims and patients with bipolar disorder. Am J Psychiatry 1997;154:1148–50.
38. Minozzi M, Nordio M, Pajalich R. The combined therapy myo-inositol plus D-chiro-inositol, in a physiological ratio, reduces the cardiovascular risk by improving the lipid profile in PCOS patients. Eur Rev Med Pharmacol Sci 2013;17:537–40.
39. Zhu X, Schuff N, Kornak J, Soher B, Yaffe K, Kramer JH, et al. Effects of Alzheimer disease on fronto-parietal brain N-acetyl aspartate and myo-inositol using magnetic resonance spectroscopic imaging. Alzheimer Dis Assoc Disord 2006;20:77–85.
40. Chhetri DR. Myo-inositol and its derivatives: their emerging role in the treatment of human diseases. Front Pharmacol 2019;10:1172.
41. Dharnidharka VR, Kwon C, Stevens G. Serum cystatin C is superior to serum creatinine as a marker of kidney function: a meta-analysis. Am J Kidney Dis 2002;40:221–6.
42. Tiao JY-H, Semmens JB, Masarei JRL, Lawrence-Brown MMD. The effect of age on serum creatinine levels in an aging population: relevance to vascular surgery. Cardiovasc Surg 2002;10:445–51.
43. Salive ME, Jones CA, Guralnik JM, Agodoa LY, Pahor M, Wallace RB. Serum creatinine levels in older adults: relationship with health status and medications. Age Ageing 1995;24:142–50.
44. Yim J, Son N-H, Kyong T, Park Y, Kim J-H. Muscle mass has a greater impact on serum creatinine levels in older males than in females. Heliyon 2023;9:e21866.
45. Fehrman-Ekholm I, Skeppholm L. Renal function in the elderly (>70 years old) measured by means of iohexol clearance, serum creatinine, serum urea and estimated clearance. Scand J Urol Nephrol 2004;38:73–7.